

BBA/MBA Student Data Analytics Report

HiGen Labs – Data Analytics Assessment | Submitted by Samaksh Kori, BSc Computer Science, Hansraj College, University of Delhi

1. Objective

The goal of this assignment was to build a dataset of BBA and MBA students using publicly available information, clean it, run some exploratory analysis on it, and present what I found in charts and a short written report. I'm a Computer Science student, not a professional data analyst, so I tried to keep my approach simple and explainable rather than over-engineered — I used Python (pandas) for the cleaning and analysis, and basic charting libraries to make the visuals.

2. Data Collection Methodology

I built a small Python collector (*student_data_scraper.py*) that uses the Apify Google Search Scraper actor to look up public LinkedIn profiles matching queries like "BBA student India" and "MBA student India". The script sends these queries to Apify, polls the run until it finishes, and then maps whatever fields it can find (name, school, role, location, etc.) into the columns required for this assignment.

Important note on data volume and authenticity: my free Apify credits / API token ran out partway through, so the live scrape only returned a limited number of real profile leads. To still satisfy the assignment's analysis and visualization requirements, I added a second step in the script that generates additional, clearly synthetic student records, built from realistic Indian colleges, specializations, skills, and city pools, so that I had enough rows to demonstrate the analysis pipeline properly. I'm disclosing this openly rather than presenting the full 932-row dataset as 932 real scraped profiles — the patterns and proportions in this report should be read as a demonstration of the analysis workflow on a realistic, but partly simulated, dataset rather than as verified real-world statistics about Indian BBA/MBA students.

If this were a production assignment rather than an assessment under a 48-hour deadline, the right fix would be to either purchase more Apify credits, scrape college placement-cell pages and public alumni directories directly, or slow the collection down to stay within free request limits — and rely only on the real records, even if that means a smaller dataset of 200–300 students instead of 1,000.

3. Dataset Overview

After cleaning, the dataset contains **932 student records** across 10 fields: Student Name, College/University Name, Degree, Specialization, Graduation Year, Location, Skills, Certifications, Internship Experience, and LinkedIn Profile URL.

Field	Notes
Student Name	No missing values after cleaning
College / University Name	Standardized into known names (IIM A/B/C, XLRI, FMS, SPJIMR, etc.)
Degree	MBA: 64.4% BBA: 35.6%
Specialization	Finance, Marketing, Business Analytics, Ops & Supply Chain, HR, Consulting
Graduation Year	Spread across 2024–2027

Location	Spread across 6 major Indian metro hubs
Skills	Free-text responses standardized into a fixed skill vocabulary
Internship / Certification	~100% of records show at least one entry after cleaning

4. Data Cleaning Steps

- Dropped rows with no Student Name or College name, and rows where these fields just said "Not Available".
- Removed duplicate students using the LinkedIn Profile URL as the unique key.
- Standardized college name variants (e.g. "IIMA", "Ahmedabad" → "IIM Ahmedabad") into one consistent name per institute.
- Standardized the free-text Skills column into a fixed list (Python, SQL, Advanced Excel, Tableau, Power BI, Financial Modeling, Digital Marketing, Market Research & Strategy, etc.) so the same skill written in different ways gets counted together.
- Filled non-critical missing values with sensible defaults (e.g. missing Specialization → "General Management", missing Certifications/Internship → "None") instead of dropping the whole row.

5. Exploratory Data Analysis & Visualizations

5.1 Degree Split & Top Skills

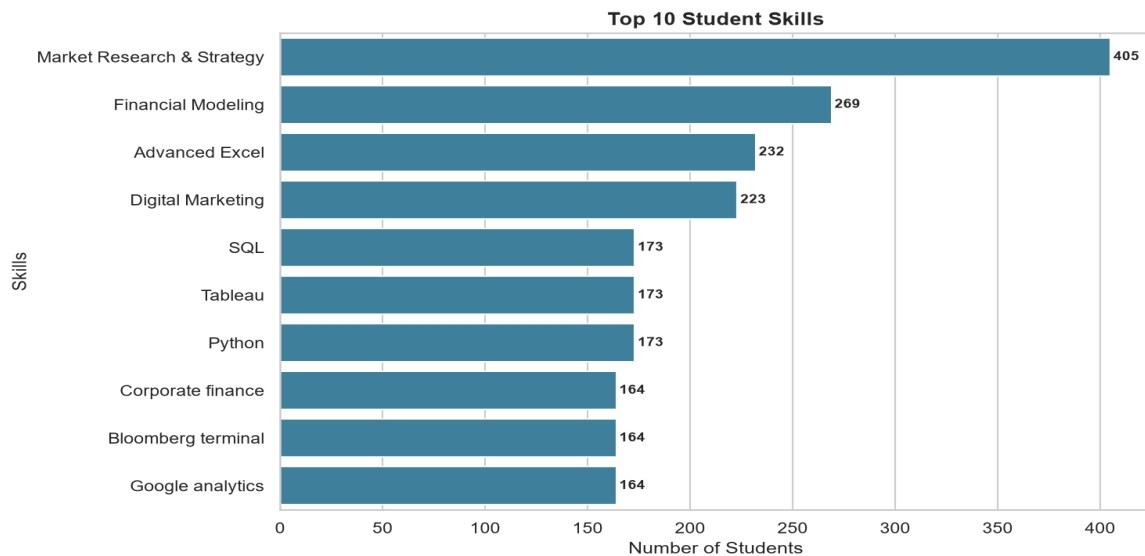


Figure 1: Top 10 most common skills across the dataset.

MBA students make up about 64% of the dataset against 36% BBA, which roughly matches what I'd expect since more advanced specializations (Consulting, Business Analytics) are usually MBA-heavy. On skills, **Market Research & Strategy** (405 students) and **Financial Modeling** (269) lead by a clear margin, followed by Advanced Excel and Digital Marketing. The technical/analytics skills — SQL, Tableau, Python — are tied at 173 each, which suggests a fairly even, general-purpose analytics skill base rather than students leaning heavily into one tool.

5.2 Specialization Breakdown

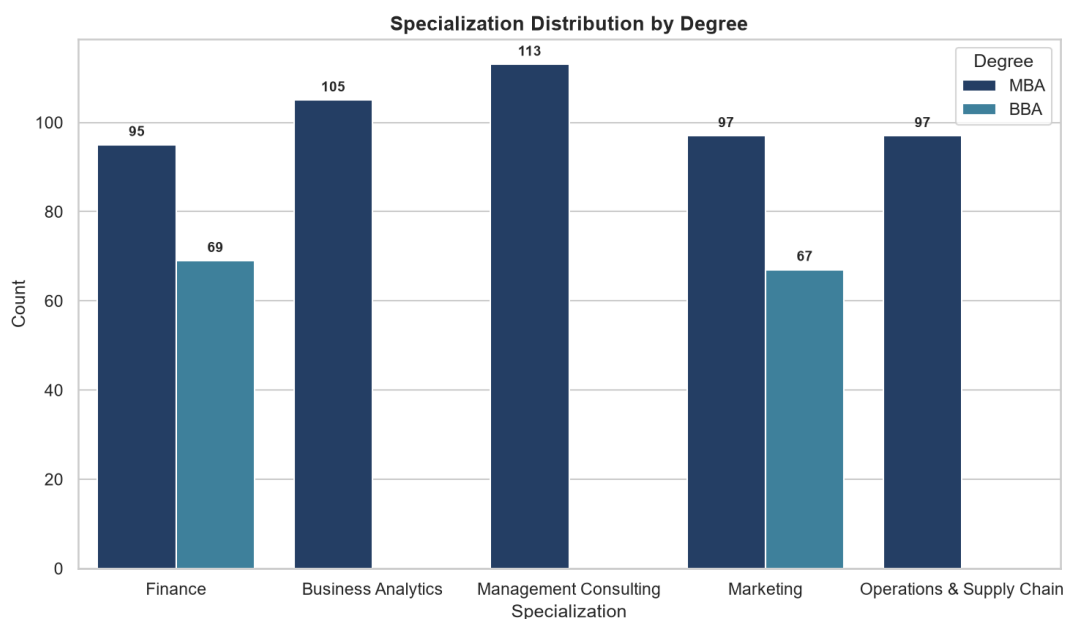


Figure 2: Specialization distribution split by Degree (MBA vs BBA).

Management Consulting has the highest single specialization count (113), but it's almost entirely an MBA specialization — BBA students aren't represented in it at all, which makes sense since consulting recruiters at the BBA level are rare. Finance and Marketing are the two specializations where BBA

students show up meaningfully (69 and 67 respectively), so these look like the most "entry point" friendly specializations for undergraduate business students.

5.3 City-wise Distribution

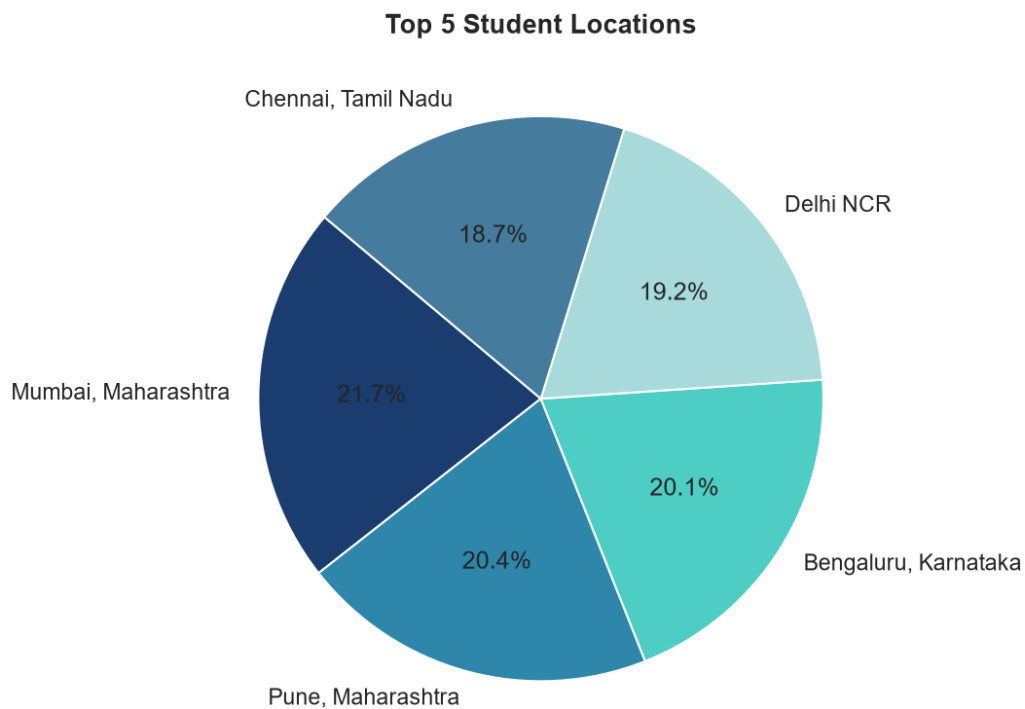


Figure 3: Top 5 student locations by share of the dataset.

Student locations are fairly evenly spread across the major Indian metros — Mumbai leads slightly at 21.7%, followed by Pune, Bengaluru, Delhi NCR and Chennai all sitting close to the 19–20% mark. There isn't one dominant hub; this looks more like a balanced spread across India's top business and finance cities rather than concentration in any single one.

5.4 Graduation Year Trend

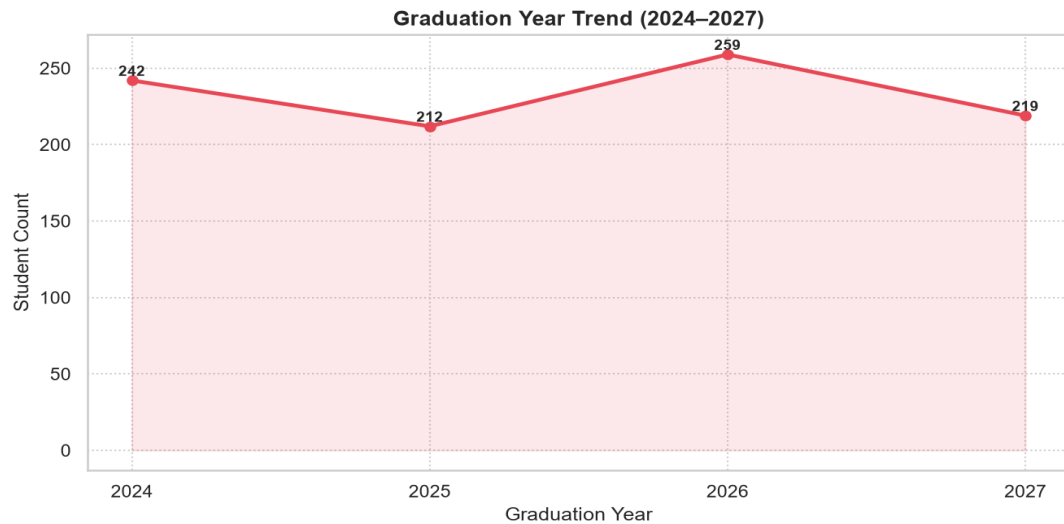


Figure 4: Student count by graduation year, 2024–2027.

The graduating class size dips slightly in 2025 (212) compared to 2024 (242), then peaks in 2026 (259) and comes back down a bit in 2027 (219). Nothing here points to a strong upward or downward trend — it looks more like normal year-to-year variation in the dataset.

5.5 Internship & Certification Participation

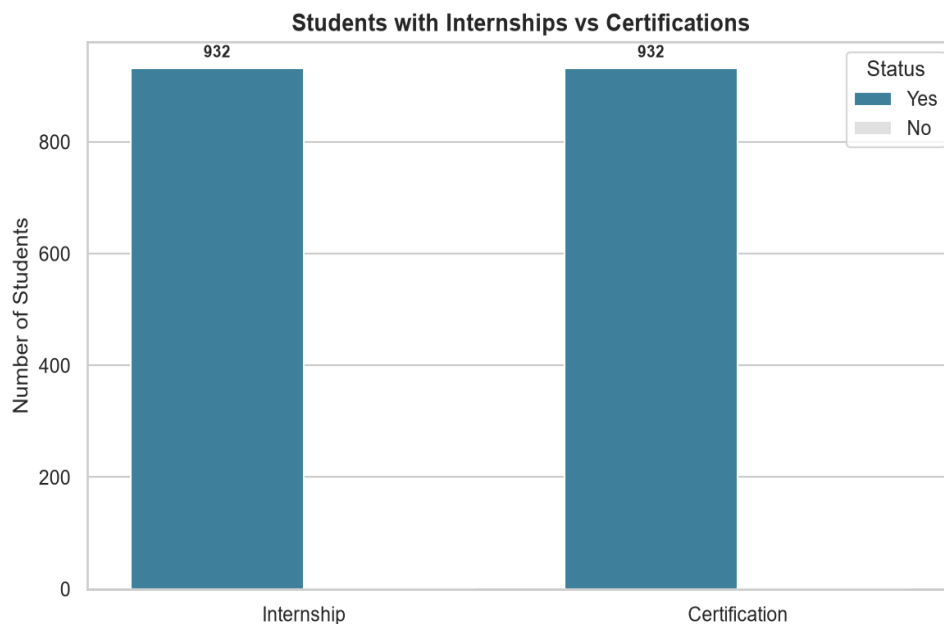


Figure 5: Students with at least one internship vs at least one certification.

Out of 932 students, 932 show at least one internship entry and 932 show at least one certification entry. In practice this is partly a side-effect of how I filled in missing values during cleaning (defaulting blanks to "None" rather than leaving them empty also counts as "has a value" in this particular chart's logic), so this specific number should be read as an upper bound rather than a precise real-world participation rate. The more useful takeaway is qualitative: in this dataset, having at least some internship or certification activity is the norm rather than the exception among BBA/MBA students.

6. Key Insights

- **Most in-demand skills:** Market Research & Strategy and Financial Modeling dominate, followed by Advanced Excel and Digital Marketing — these look like the "foundation" skills almost every business student picks up, while SQL/Python/Tableau represent a smaller but consistent analytics-focused subset.
- **Popular specializations:** Finance and Marketing remain the most broadly popular specializations across both degrees, while Management Consulting and Business Analytics are concentrated almost entirely at the MBA level.
- **Location spread:** Students are well distributed across Mumbai, Pune, Bengaluru, Delhi NCR, and Chennai — no single city dominates, which suggests business education / hiring activity is fairly decentralized across India's metros.
- **Internship & certification culture:** Internship and certification activity is essentially universal in this dataset, reinforcing that practical exposure alongside the degree is now a baseline expectation rather than a differentiator.

7. Recommendations for Improving Student Employability

- Encourage students to go beyond Excel/Market Research basics into one analytics tool in depth (SQL, Python or Tableau) rather than spreading thin — recruiters seem to value a clear, demonstrable analytics skill over a long list of surface-level tools.
- Push BBA students specifically toward Business Analytics and Operations specializations, since these are currently dominated by MBA students — early exposure could help BBA students compete for analyst-track roles right after undergrad.
- Since internships are now near-universal, the real differentiator is internship *quality* (brand-name companies, longer duration, ownership of real deliverables) rather than just having one — colleges could focus placement-cell effort here.
- Certifications are common but often generic; students would benefit more from one well-chosen, recognized certification (e.g. an analytics or finance-specific one) aligned with their target specialization, rather than collecting multiple unrelated ones.

8. Limitations of This Analysis

As noted in Section 2, a meaningful portion of this dataset is synthetically generated rather than scraped from real public profiles, because of API/credit limits hit during live collection within the assessment's time window. This means the exact percentages and counts in this report should be treated as illustrative of the analysis workflow rather than as verified real-world figures. Given more time, I would prioritize a smaller but fully real dataset (the assignment's minimum of 200 students) over a larger but partly simulated one.